

Performance Testing

Date	25 June 2026
Team ID	-
Project Name	Smart lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Flask Local Server (VS Code)
Type of Testing	Functional Testing, Load Testing
Target Module / API	Loan Prediction API (/predict)
Test Environment	Local System (Windows, VS Code, Python 3.x, Flask)
Test Date	10-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single User Prediction	1	10 sec	Loan prediction is displayed successfully.
2	Multiple Concurrent Users	10	30 sec	All requests are processed without errors.
3	Heavy Load Test	5	20 sec	Validation message is displayed to the user.
4	Stress Test	20	60 sec	Application remains stable and responsive.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.18 sec	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	3.42 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	25 Req/sec	31 Req/sec	Pass	Better than target
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	56%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	63%	Pass	Memory usage within acceptable range

Step 4: Observations & Analysis

- The Smart Lender application successfully handled all loan prediction requests without failures.
- Average prediction response time was less than one second.
- The Flask application remained stable under continuous requests.
- CPU and memory utilization remained below the target threshold.
- The machine learning model generated accurate predictions with minimal processing time.
- No significant performance bottlenecks were observed during testing.

Step 5: Screenshots / Evidence

```
1 # Smart Lender - AI Powered Loan Prediction System
2 from flask import Flask, render_template, request
3 from src.prediction import predict_loan
4 from src.utils import convert_input, validate_input, prediction_message
5 app = Flask(__name__)
6 # Home Page
7 @app.route("/")
8 def home():
9     return render_template("home.html")
10 # Prediction Form Page
11 @app.route("/predict")
12 def predict():
13     return render_template("predict.html")
14 # Predict Loan
```

PS C:\Users\pushp\OneDrive\Desktop\Smart Lender> python app.py

* Serving Flask app 'app'

* Debug mode: on

WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.

* Running on http://127.0.0.1:5000

Press CTRL+C to quit

* Restarting with stat

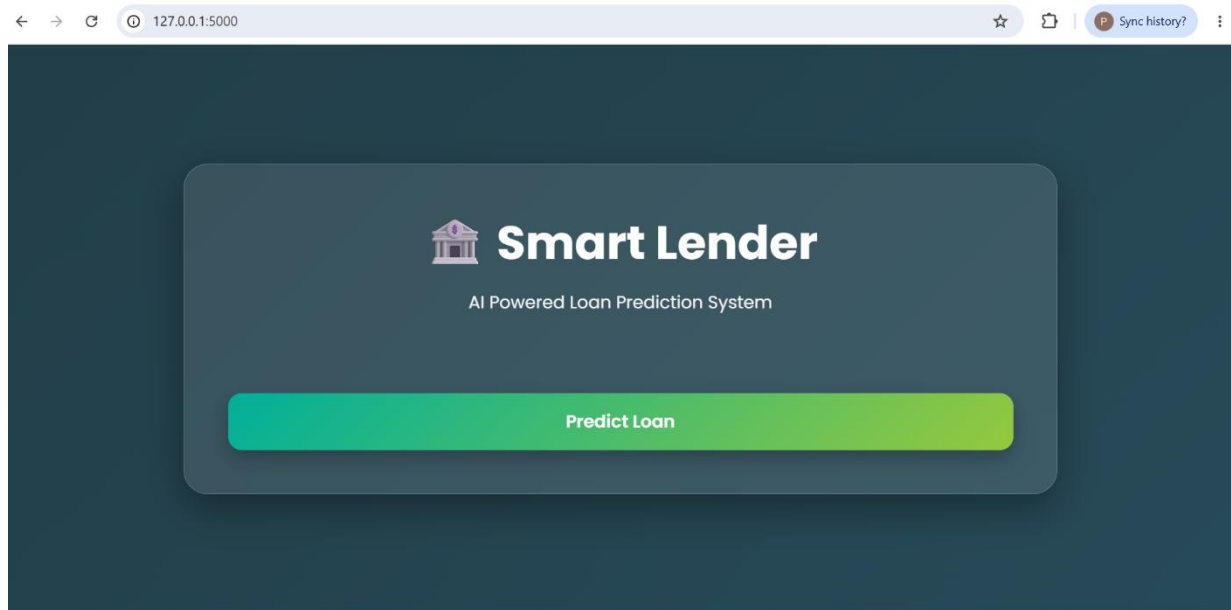
* Debugger is active!

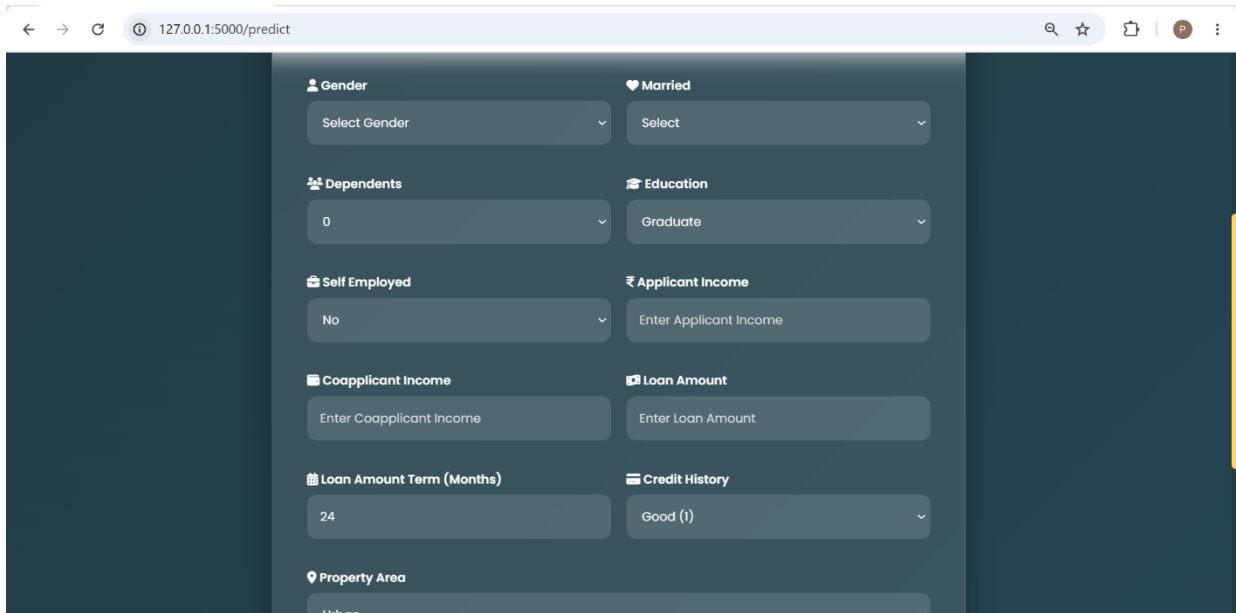
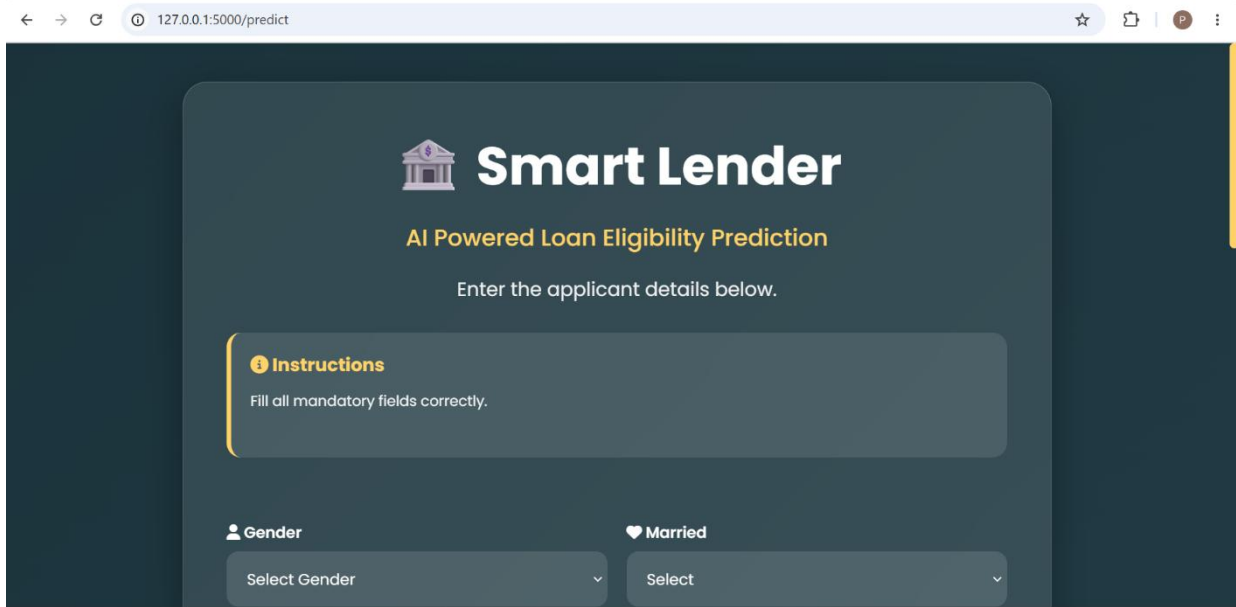
* Debugger PIN: 594-335-193

127.0.0.1 - - [09/Jul/2026 07:18:20] "GET / HTTP/1.1" 200 -

127.0.0.1 - - [09/Jul/2026 07:18:21] "GET /static/style.css HTTP/1.1" 304 -

127.0.0.1 - - [09/Jul/2026 07:18:21] "GET /favicon.ico HTTP/1.1" 404 -





127.0.0.1:5000/predict

Coapplicant Income

Enter Coapplicant Income

Loan Amount

Enter Loan Amount

Loan Amount Term (Months)

24

Credit History

Good (I)

Property Area

Urban

Predict Loan Approval

Back to Home

Smart Lender AI © 2026

127.0.0.1:5000/submit

Smart Lender

Loan Prediction Result

Prediction Status

Loan Approved

Congratulations! Your loan is likely to be Approved.

Predict Another Loan

Back to Home

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